UMHackthon 2025

Project Title:

MEX Assistant – Insights Proposal

Team Name:

NEURO STORM

Team Members:

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1.0 Executive Summary

We propose MEX Assistant – Insights, an AI-powered, chat-based assistant designed to empower Grab's merchant-partners with real-time business analytics, personalized guidance, and multilingual support. The assistant delivers timely alerts, sales trends, inventory warnings, and competitive benchmarking, helping merchants make fast, informed decisions and grow their businesses efficiently.

2.0. Problem Understanding

Grab's merchant-partners face several key challenges:

2.1. Difficulty identifying actionable trends quickly

2.2. Lack of personalized guidance

2.3. Language and tech literacy barriers

Our solution tackles these issues through real-time, personalized, and easy-to-understand AI interactions. The assistant provides immediate insights into sales trends, inventory levels, and operational performance, enabling merchants to make swift, informed decisions. It also provides personalized tips and strategies to support their growth and improvement by analysing factors such as merchant type, location, and size. Multilingual support ensures that the assistant communicates in multiple languages using simple, clear messages to guarantee all merchants can easily understand and act on the information.

3.0. System Architecture

3.1. Data Sources:

Items.csv

No	Column Name	Data Type
1.	item_id	INTEGER
2.	cuisine_tag	STRING
3.	item_name	STRING
4.	item_price	FLOAT
5.	merchant_id	STRING

Merchant.csv

No	Column Name	Data Type
1.	merchant_id	STRING
2.	merchant_name	STRING
3.	join_date	INTEGER
4.	city_id	INTEGER

Keyword.csv

No	Column Name	Data Type
1.	keyword	STRING
2.	view	INTEGER

3.	menu	INTEGER
4.	checkout	INTEGER
5.	order	INTEGER

Transaction_data.csv

No	**Column Name**	**Data Type**
1.	order_id	STRING
2.	order_time	DATETIME
3.	driver_arrival_time	DATETIME
4.	driver_pickup_time	DATETIME
5.	delivery_time	DATETIME
6.	order_value	FLOAT
7.	eater_id	INTEGER
8.	merchant_id	STRING

Transaction_items.csv

No	**Column Name**	**Data Type**
1.	order_id	STRING
2.	item_id	INTEGER
3.	merchant_id	STRING

3.2. Backend Services:

Machine Learning Solution

No	Goal	Service	ML Technique	Algorithms	Dataset
1	Forecast future sales trends to help merchants prepare inventory and plan promotions.	Sales trend prediction	Time Series Forecasting	ARIMA, Prophet, LSTM	Transaction_data, Transaction_items, Items
2	Show top-performing products to merchants and recommend similar items.	Top product insights	Descriptive Analysis, Clustering	KMeans, DBSCAN	Transaction_items, Items
3	Analyze which keywords lead to successful orders (from view -> checkout -> order)	Keyword conversion	Conversion Rate Analysis, Feature Importance or Regression	Logistic Regression, Decision Trees, Gradient Boosting (e.g.XGBoost)	Keyword
4	Detect sudden drops or spikes in sales to alert merchants.	Anomaly Detection for Sudden Sales Drop	Anomaly Detection	Isolation Forest, One-Class SVM,	Transaction_data

				Autoencoders (if using deep learning)	
5	Analyze customer reviews and generate sentiment scores	Feedback Analysis	Sentiment Analysis with NLP	Logistic Regression /SVM /BERT /VADER	Add-on with Customer Reviews Data
6	Recommend which items to promote or bundle based on customer purchase patterns	Personalized Recommendations for Merchants	ARM	Apriori, FP-Growth	Transaction_items
7	Compare a merchant's performance with others in the same city or cuisine.	Merchant Performance Benchmarking	Ranking + Clustering	KMeans, Ranking Methods (e.g. percentile ranking)	Merchant, Transaction_data
8	Alert if items are selling faster than usual (stock might run out)	Inventory alert System (Simulated with Sales Drop or Spike)	Forecasting	Regression	Transaction_items

9	Recommend best practices or expected sales for newly joined merchants	New Merchant Onboarding Insights	Predictive Modeling + Clustering	Random Forest/ Decision Trees/ Regression for revenue prediction	Merchant, Compare with others' past onboarding performanc e.

Alert generation

- Send notifications about the merchandise that is nearly out of stock.
- Send notifications about current high sales products for restocking purposes.

Tools and Action needed

- REST API
- Send HTTP request
- Backend respond with JSON file

3.3. NLP Engine

Able to extract information from the dashboard.

- Pre-processing
  - Normalize repetitive characters
  - Convert to lower case
  - Remove punctuation and numbers
  - Correct spelling
  - Tokenization
  - Remove stopwords
  - Lemmatization
- Feature Extraction
  - Spacy Embeddings
  - Synonyms
- Algorithm Used
  - Rule Based Method (Free of charge)
  - Generative Chatbots (Not Free)

    - Cloud-hosted LLMs like GPT-4, Claude, and Gemini Pro

    - Open-Source Generative Models like LLaMA3 and Mistral need GPU server to run on cloud or local.

3.4. Frontend

The frontend of the application features a mobile-friendly chatbot interface that allows merchant-partners to interact with the AI assistant for insights and support, alongside a dynamic sales dashboard that visualizes key business metrics such as sales trends, top-selling items, and customer feedback analysis. Developed using HTML, CSS, and JavaScript, the interface ensures a responsive and user-friendly experience, while Chart.js is utilized to create interactive charts and graphs that help merchants easily interpret and act on real-time data insights.

4.0. Core Features

4.1. Real-Time Business Insights

The MEX Assistant's real-time business insights module is designed to transform raw transaction data into immediately actionable intelligence by allowing detect sales drops or spikes, inventory shortage, and monitoring peak sales hour identification. All of the activities of business will always be in real-time processing where the system is able to detect the minor anomalies changes of the business in advance. Thus, sellers can view the recommendations given by the system and also read the customer reviews to take further action to solve those problems in order to make the enhancement. Other than that, merchandise can read customer reviews

4.2. Personalized Merchant Guidance

The system delivers personalized, actionable recommendations tailored to each merchant's specific characteristics, such as business type, regional market, and store size. These insights help identify growth opportunities and suggest concrete improvements to enhance performance. Merchants receive timely alerts for products that are nearly out of stock, along with notifications for high-demand items, enabling proactive restocking and demand management. The system also offers product pairing suggestions to increase average cart size and provides promotional timing insights based on seasonal trends and customer behavior. Additionally, merchants benefit from market comparisons, including competitive benchmarking, emerging trend insights, and category performance analysis.

4.3. Intuitive Communication

The system supports intuitive communication by incorporating multiple features. Firstly, it provides multilingual support, allowing users to interact in English, Malay, or Mandarin, thereby catering to the diverse language preferences in Southeast Asia. Additionally, it delivers information through visual summaries, such as charts and graphs, which help users quickly grasp key insights. The dialogue tone is designed to be colloquial and friendly, making the assistant feel more approachable and natural. This is achieved through a well-structured intents JSON file, which includes fields such as tags, patterns, and responses. These elements enable the system to respond with relevant and context-aware answers to customer queries, ensuring smooth and meaningful conversations.

5.0. Personalization Strategy (Data-driven)

To deliver relevant and actionable insights, the AI assistant generates personalized recommendations and guidance for each merchant by leveraging available data fields such as transactional, behavioral, and profile data. The insights span merchant operation, customer pattern, and performance across several dimensions.

5.1. Merchant category

Data used: Item metadata (*cuisine_tag, item name*), transaction patterns, customer purchase behaviour

The AI assistant identifies **merchant types** by using **cuisine tags** or **menu item keywords**. It analyze customers purchasing behaviour to uncover trends such as:

- Where customers frequently buy from

- Whether customers often purchase similar items from different merchants
- Customer spending habits and loyalty indicators

By analyzing these behaviour, AI assistant can:

- Infer competitive sets by clustering merchants with similar offerings and overlapping customers bases
- Recommend product expansion such as “Add drinks to bundle with your top selling item”
- Suggest targeted promotions for customer retention, e.g., loyalty rewards, discounts for returning customers

Example AI suggestions:

- “Many of your customers also order **milk tea** from nearby shops. Consider adding beverages to your menu to increase cart value”
- “Customers tend to spend between RM10-12 per order. You may offering a bundle within this range could boost conversions”

5.2. Location Based Personalization

Data used: *city_id*, item pricing, sales volume, popular items in region

The AI assistant performs city-based benchmarking by analyzing:

- Price ranges across item categories
- Popular items in the city
- Sales trends within similar merchants

Based on the result of data analyzation, the AI assistant recommends:

- Price adjustments to remain competitive
- Menu updates by adding high-demand items missing from the menu
- Localized promotions targeting customer preferences in that city

Example AI suggestions:

- “In cities like City 4, chicken rice bowls priced between RM8 - 10 perform best. Consider adjusting your price range”
- “Cheesy fries are among the top-selling side items in your city but are not currently on your menu”
- “Sales in cities like City 2 peak during weekday lunch hours. A ‘Lunch Set’ promotion from Mon–Fri could increase orders.”

5.3. Business Maturity (joining date and merchant performance)

Data used: join_date, merchant performance metrics over time

The AI assistant will provide recommendations based on merchant maturity, which categorized as either newly onboarded or long-standing merchants.

For newer merchants:

- Onboarding guidance and setup tips
- Pricing strategy
- Photo and menu optimization tips
- Visibility improvements to attract customers.

For established merchants:

- Deeper analysis into historical sales and trend
- A/B testing ideas for promotions
- Menu diversification and expansion strategies

Example AI suggestion:

- “As a new merchant, consider starting with high-quality photos and 3 best-selling bundles to attract early buyers.”
- “Your top item ‘Spicy Wings’ consistently sells well—test bundling it with a drink for added value.”
- “Your historical data shows a dip in January—consider running a New Year campaign to counter the trend.”

5.4 Operation Strategy (Delivery Time Analysis)

Data used: *driver_arrival_time*, *delivery_time*, average prepare time, order volume, customers' ratings

Delivery performance is crucial to customer satisfaction and retention, the AI assistant analyzes:

- Average prepare and delivery time
- The relationship between customer rating and delivery performance

Then AI assistant recommends strategies on:

- Preparation time
- Adjusting delivery radius
- Prioritizing faster preparation items during peak hours

Example AI Suggestion:

- "Your average delivery time is 42 minutes, which is 15% above the city average. Consider reducing prep time or limiting delivery radius during peak hours."
- "Highlight fast-prep items like 'Chicken Bites' for dinner rush hours to speed up turnaround."
- "Orders with longer delivery times tend to receive lower ratings. Improve operational flow to prevent churn."

5.5. Personalized Approaches

To generate intelligent and tailored recommendations, the AI assistant use two complementary methods which are rule-based logic and machine learning. These approaches are powered by structured data integration using SQL queries, linking merchant, customer, item and transaction data.

5.5.1 Rule-based logic

Initial insights are driven by straightforward conditions and business rules. For example:

- Time based promotions. If peak orders occur during weekends, recommend promoting on Fridays.
- Item-level suggestions. If item prices are significantly higher than local competitors, suggest adjusting price or bundling with value items.
- If customer often reorder the same item, recommend creating “customer favourite” category

This approach provides fast, interpretable insights especially effective for early-stage personalization or for merchants with limited data history.

5.5.2 SQL-based Data Linking

The foundation of the personalization engine relies on SQL-style data joining across multiple datasets to extract insights:

Data Flow Example that shows customer behaviour analysis:

1. Identify merchant’s customer

   Use *merchant_id* from *Transaction Data* to extract unique *eater_id.*

2. Cross-merchant behaviour

   Use *eater_id* in full *Transaction Data* to trace customer order across different merchants.

3. Compare purchase behaviour

   Join with *Item Data*, *Merchant Data* to identify similarities or gaps in pricing, category, item types.

4. Behavioural insight
    - Are customers price-sensitive?
    - Are they loyal to this merchant?
    - Do they order similar food elsewhere?
5. Strategies suggestion
    - Loyalty programs to retain shared customers
    - Bundles that align with competitors' strong-selling categories
    - Adjust pricing if customer spending behavior differs across merchants
    -

5.5.3. ML-Based clustering

Beyond rule-based and SQL analysis, AI assistant applies clustering techniques to identify merchants with similar pattern across to perform:

- Clustering of merchants by similar behavior, menu offerings, or customer base
- Customer segmentation based on order frequency, spend level, and retention risk
- Recommendation generation for promotions or pricing based on top-performing peers

Unlike rule-based logic or SQL joins, ML clustering surfaces hidden patterns at scale—especially useful when behavior is non-linear or when dealing with sparse datasets.

6.0. Conversational Design Example flows

1. **Greeting**
    - Welcoming the user and starting a friendly interaction.

2. **Business Summary**
    - Providing an overview or summary of the merchant's overall business performance.

3. **Top Products**
    - Showing best-selling or top-performing menu items.

4. **Customer Feedback Summary**
    - Summarizing overall customer sentiment or reviews.

5. **Negative Feedback Details**
    - Highlighting issues or complaints from customers.

6. **Improvement Tips**
    - Giving general business or operational suggestions for improvement.

7. **Promotion Ideas**
    - Offering promotional or campaign suggestions to boost sales.

8. **Delivery Issues**
    - Addressing problems with delivery times or customer complaints about delivery.

9. **Product Quality**
    - Providing tips to improve food/item quality based on feedback.

10. **Merchant Profile**

- Displaying the merchant's own store/account/profile information.

11. **Thank You**

- Handling gratitude from the user and closing conversations politely.

7.0. Prototype Showcase

Alice, is a cake shop owner in Malaysia. Today, she wants to use MEX Assistant to manage her business better. She will interact with the assistant to manage the inventory, tracking her sales and some decision-making under pressure.

Our MEX Assistant will help her:

7.1. Real-time alerting

- ❖ Demo Chat Example:

[Alert: Hi Alice's Cafe! Your Chocolate Brownies are selling out fast(only 10 left)!
Want to restock now? >Restock >View Inventory]

This short message ca[n let Alice immediately realize that is the time to restock the product before the situation is out of stock.
The notification also comes with easy action buttons which are [Restock] and [View Inventory].
It is based on real-time data like live sales trends.

7.2. Contextual tips

- ❖ Demo Chat Example:

[Tip: “Try making a meal set of chocolate brownies with smoothies to boost afternoon sales by 20%.]

The MEX Assistant doesn’t just alert, it gives smart business advice intelligently. It can personalized the tips based on live merchant data to help Alice act proactively.

7.3. Visual insight rendering

- ❖ Demo Visual:

[“Customer Feedback Overview”]
[Mini Bar Chart Image]
[--------------------------68%   positive]
[--------------22%   neutral]
[----10%   negative]

MEX Assistant generates a simple chart for fast understanding. It only shows easy summaries and not technical jargon. This can help Alice easily have an overview about customer feedback of the cake business.

7.4. Multilingual chatbot conversations

- ❖ Demo Chat Example:

[“Halo Alice! Stok untuk Chocolate Brownies tinggal sedikit (15 porsi). Mau buat promosi tambahan untuk habiskan stok ke?”]

Alice can change the default language which is English to Malay which is more friendly to her. The assistant also shows localized expressions so it keeps merchants comfortable and confident in their preferred language.

8.0. Evaluation Alignment

8.1. Insight Quality and Relevance

In our prototype, it has actionability and prioritization. MEX Assistant gives us immediate and clear actions offered after every alert or tip. For example, it [suggest restock], [create meal] and [make promotion Buy 1 Free 1] without needing extra analysis.

MEX Assistant makes the alert only for the important trends of business. For instance, it shows an alert notification about one of the selling products running out of stock, not every small data change.

8.2. Communication Effectiveness

Based on our prototype, we can find that the MEX Assistant is using simple language like “Low Stock Alert”, “68% positive”(for customer feedback) without jargon or technical words.

The assistant also uses the graphs to show the sales trend by date period like real-time, yesterday, past 7 days, past 30 days, by day, by week or by year. This can let mechant easily overview the sales without counting by manual. It also uses colour indicators like green colour for increase, orange colour for remaining unchanged and red colour for decrease. The purpose is to make the insights instantly understandable.

8.3. User Experience and Engagement

Following our prototype, we can see that MEX Assistant has a fluent chat like it will alert merchants that the stock is low, then it gives suggestions for merchants to restock it. After that, the merchant will ask the assistant whether she wants to restock it. If yes, then merchants only need to make an easy decision by clicking the restock button. It is not a command but is a suggestion.

MEX Assistant also has to respond in chat very fast within 1 to 2 seconds for insights after an event happens. For example, merchant is requesting to see the shop performance in real-time, then the assistant has to give an instant response that shows the line graph of the sales in real-time. Another example is when assistant finds that the product is running out of stock, then the merchant will get a notification that the stock is running out right away.

9.0. Future Improvements

9.1. Cross-platform integration

Deploy the analytics widgets within the existing Grab dashboard. Sellers are able to view the information of metrics and receive the alerts from the system without leaving their workflow on the current time. The lightweight interactive dashboards will prompt out the alerts that need to be caution by sellers. To solve the most important alert as soon as possible, the system will prioritize the critical urgency level to make the priority. The alerts notification can be sent through the sellers' WhatsApp or even SMS. Hence, sellers can always take note of the messages of the sensitive alerts in order to get the immediate action. Other than that, deploy voice interaction functions in order to let sellers accomplish with hands-free access to insights. This can make the efficiency sellers increase since they can continue doing their workload during the peak operational hours without physically hands-on with their phone. Thus, sellers can receive the feedback and notifications through the voice assistant.

9.2. Predictive analytics and forecasting

Deploy an intelligent prediction and forecast models by incorporating weather situations, local events calendars, and restocking alerts. This enables sellers to make the planning and scheduling of the product that they want to promote or sell. For example, sellers can increase the stock of steamboat when the system notices that day will be a rainy day. Meanwhile, by integrating local event calendars, sellers can always track the activities which organize nearby their shop. Hence, they can base on this information, extend the operating hours and prepare more merchandise. Besides, the system can give the suggestion of restocking the inventory when it predicts that the merchandise will finish in a short period in order to help sellers optimize the preparation times.

10.0. Team & Tools

10.1. Tools: Python, TensorFlow, Dialogflow, JS/React, Figma

10.2. Roles: Data Engineer, AI Developer, UX Designer, Frontend Dev, Project Manager

11.0. Appendix

11.1. Sample JSON structure


```
{
  "intents": [
    {
      "tag": "greeting",
      "patterns": ["Hi", "Hello", "Good day", "Hey", "Hi there"],
      "responses": [
        "Hello partner! Ready to explore your business insights?",
        "Hey there! How can I help you grow your Grab store today?"
      ]
    },
    {
      "tag": "business_summary",
      "patterns": ["Show me the business overview", "Give me the performance summary",
"Business summary please"],
      "responses": [
        "Here’s your business summary: sales trends, best items so far.",
        "Sure! Let me pull up your business performance dashboard for a quick overview.
Click this
         link please. (Redirect to dashboard page)"
```

```
      ]
    },
    {
      "tag": "top_products",
      "patterns": ["What are my top selling items?", "Best performing menu", "Top products this week"],
      "responses": [
        "Your top performers this week are [Product A], [Product B], and [Product C]!",
        "Looks like these items are selling the most right now — keep it up!"
      ]
    },
    {
      "tag": "customer_feedback_summary",
      "patterns": ["Customer sentiment summary", "What are customers saying?", "Give me customer feedback overview"],
      "responses": [
        "Here’s a breakdown: 68% positive, 22% neutral, and 10% negative feedback from customers.",
        "Most customers are loving your service! Want to dive deeper into what they’re saying?"
      ]
```

},

{

"tag": "negative_feedback_details",

"patterns": ["Show me what went wrong", "Negative reviews", "Any complaints from customers?"],

"responses": [

"Here are the top concerns raised by your customers — let’s tackle them one by one.[concern1], [concern2], [concern3]",

"Looks like some reviews mentioned delays and item mismatch. Want some tips to fix them?"

]

},

{

"tag": "improvement_tips",

"patterns": ["How can I improve?", "Tips to increase sales", "Suggestions to improve my business"],

"responses": [

"You could try improving delivery speed, offering combos, and responding quickly to reviews.",

"Want to grow faster? Keep top items in stock and use trending promos — I can help you plan it."

]

```
  },
  {
    "tag": "promotion_ideas",
    "patterns": ["How to boost sales?", "Promotion ideas?", "Any campaign suggestions?"],
    "responses": [
      "Try a 'Buy 1 Free 1' for slow-moving items, or discounts on top sellers during lunch hours.",
      "Let’s run a campaign for your regulars — loyalty perks and flash sales work well!"
    ]
  },
  {
    "tag": "delivery_issues",
    "patterns": ["Customers complain about delivery", "Late delivery problems", "Issues with delivery time"],
    "responses": [
      "You might want to tighten prep time or check rider availability during peak hours.",
      "Delivery delays were mentioned a few times — let's look into optimizing your order handling flow."
    ]
  },
```

{

"tag": "product_quality",

"patterns": ["Customer says food is bad", "How to improve food quality?", "Low rating on product quality"],

"responses": [

"Food quality concerns came up. Maybe it’s time for a quick kitchen audit or retraining.",

"Consistent portion sizes and fresh ingredients make a big difference — need a checklist?"

]

},

{

"tag": "merchant_profile",

"patterns": [

"Show my profile",

"What is my merchant info?",

"Tell me my store details",

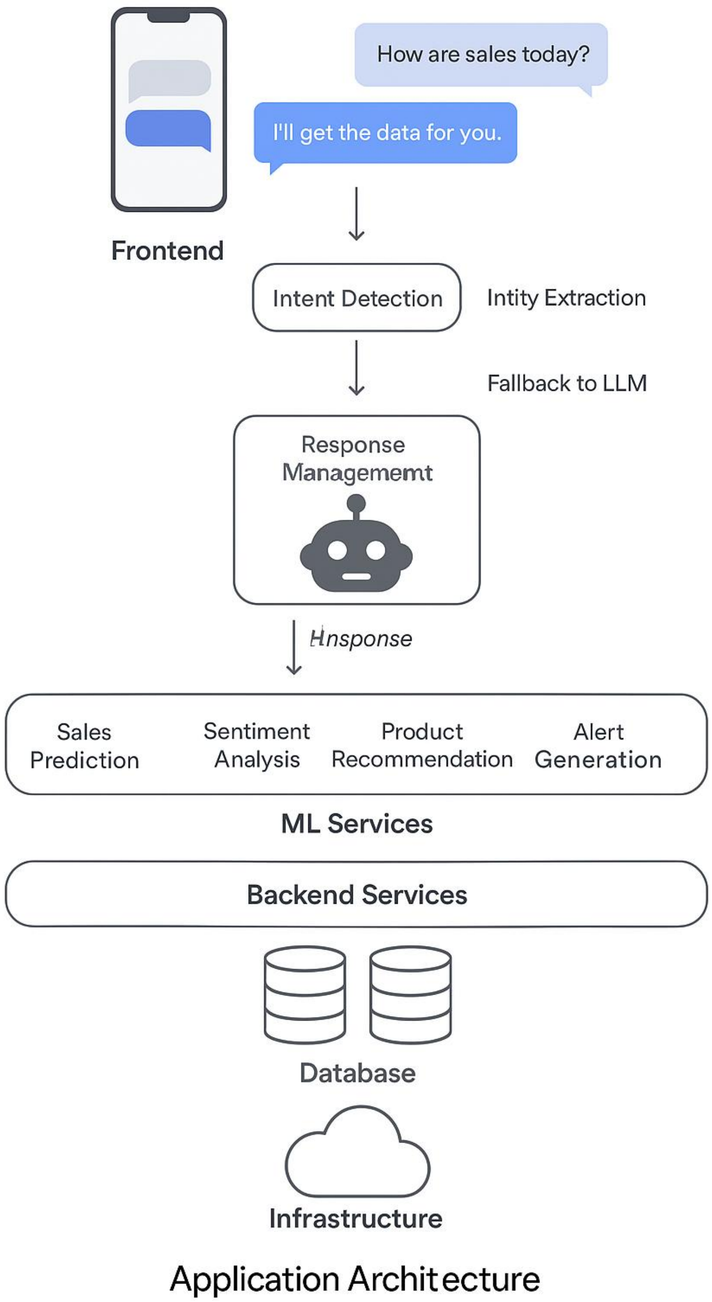
"Give me my account summary"

],

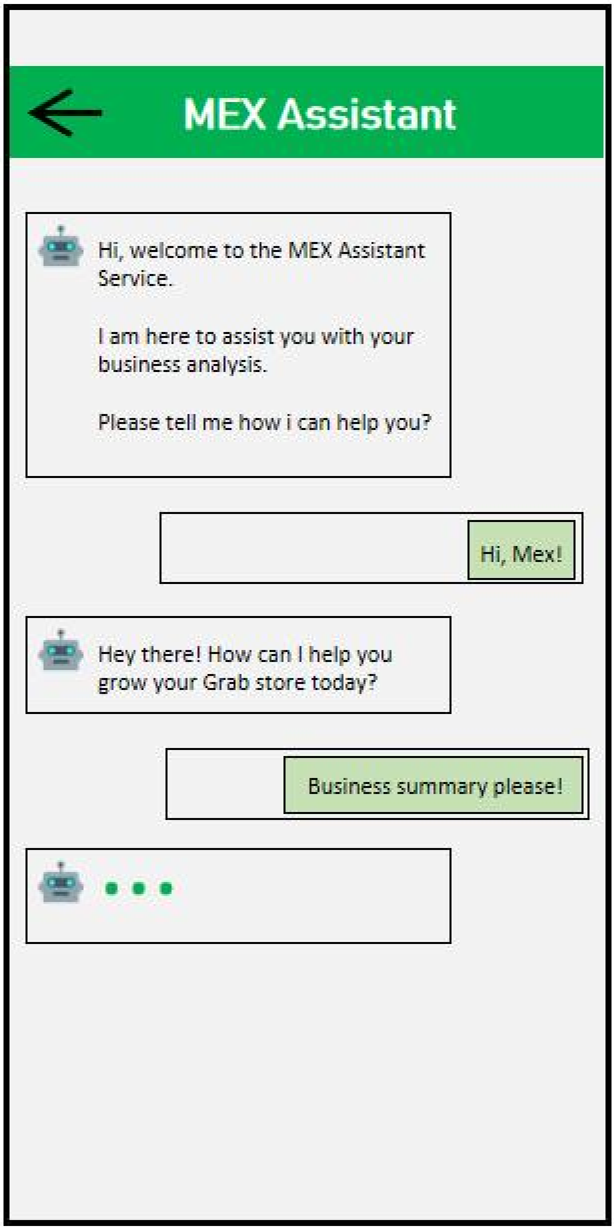
"responses": [

```
        "Here's your merchant profile: Name: [Your Name], Store: [Your Store], Location: [Your Area], Rating: [4.6⭐]",
        "Got it! You're currently registered as [Store Name] at [Location]. Want to update anything?"
      ]
    },
    {
      "tag": "thank_you",
      "patterns": ["Thanks", "Thank you", "Appreciate it", "Good support"],
      "responses": [
        "You're most welcome! Let me know if there's anything else you need.",
        "Glad I could help! Let's keep making your business stronger    "
      ]
    }
  ]
}
```

12.2. Data model schema



12.3. Dialogue examples



12.4. Screenshots of rule-based chatbot basic working prototype